LAB Assignment: Recap Neural Network Foundations including CNN (Ref My video Lectures on Introduction to Machine Learning-45-51) Course Name and instructor: GAI , Prof G C Nandi

Use MNIST database of handwritten digits, I think it is a perfect dataset for learning such concepts. Also start solving the problems in such a way using OOM so that apart form GDA,SGDA, we can implement three optimization algorithms- AdaGrad, RMSProp and ADAM on it and compare the results..

Hints:

1. Backpropagation Implementation:

- Uses a fully-connected neural network architecture
- Implements forward and backward propagation
- Includes mini-batch training
- Uses sigmoid activation function
- Implements gradient descent optimization (compare the performance of all three Batch GDA, SGDA(stochastic gradient descent algorithm),and mini-batch GDA)

2. CNN Implementation:

- Includes convolutional layers, ReLU activation, and max pooling
- Has a simple architecture: Conv -> ReLU -> MaxPool -> Conv -> ReLU -> MaxPool -> FC
- Implements basic forward pass and simplified training
- Uses MNIST dataset (28x28 grayscale images)
- For both implementations use only NumPy for numerical

  computations B.Experimentations to be evaluated after the

  theory is covered in the class:

1. Try different network architectures by changing layer sizes
2. Experiment with learning rates and batch sizes
3. Add additional features like:
   - o Different activation functions (ReLU, tanh)
   - o Dropout regularization
   - o Batch normalization
   - o Different optimization algorithms (AdaGrad, RMSprop, ADAM)

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